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Deepfake Audio Detection via MFCC Features Using Machine Learning

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Abstract

Developing lightweight audio authentication tools for resource-constrained IoT and mobile devices is critical. This study extracts Mel-Frequency Cepstral Coefficients (MFCCs) alongside delta features.

Key Methodology & Architecture

MFCC extraction, 1st and 2nd derivative deltas, Random Forest classifier, Support Vector Machine.

Datasets & Benchmark Environments

ASVspoof 2019 Physical and Logical Access Datasets.

Performance Metrics & Graph Axis Parameters

Reported Results: Random Forest Accuracy = 96.5%, Inference Speed = 2.1ms/sample, CPU RAM Usage < 45MB.

Graph Parameter Mapping: X-axis: Number of MFCC Coefficients (13, 26, 39, 52 features); Y-axis: Classification Accuracy (% from 85% to 100%).

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Ultra-lightweight computational footprint, runs on edge microcontrollers and mobile apps.	Lower robustness against heavy MP3 lossy compression compared to deep learning methods.

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